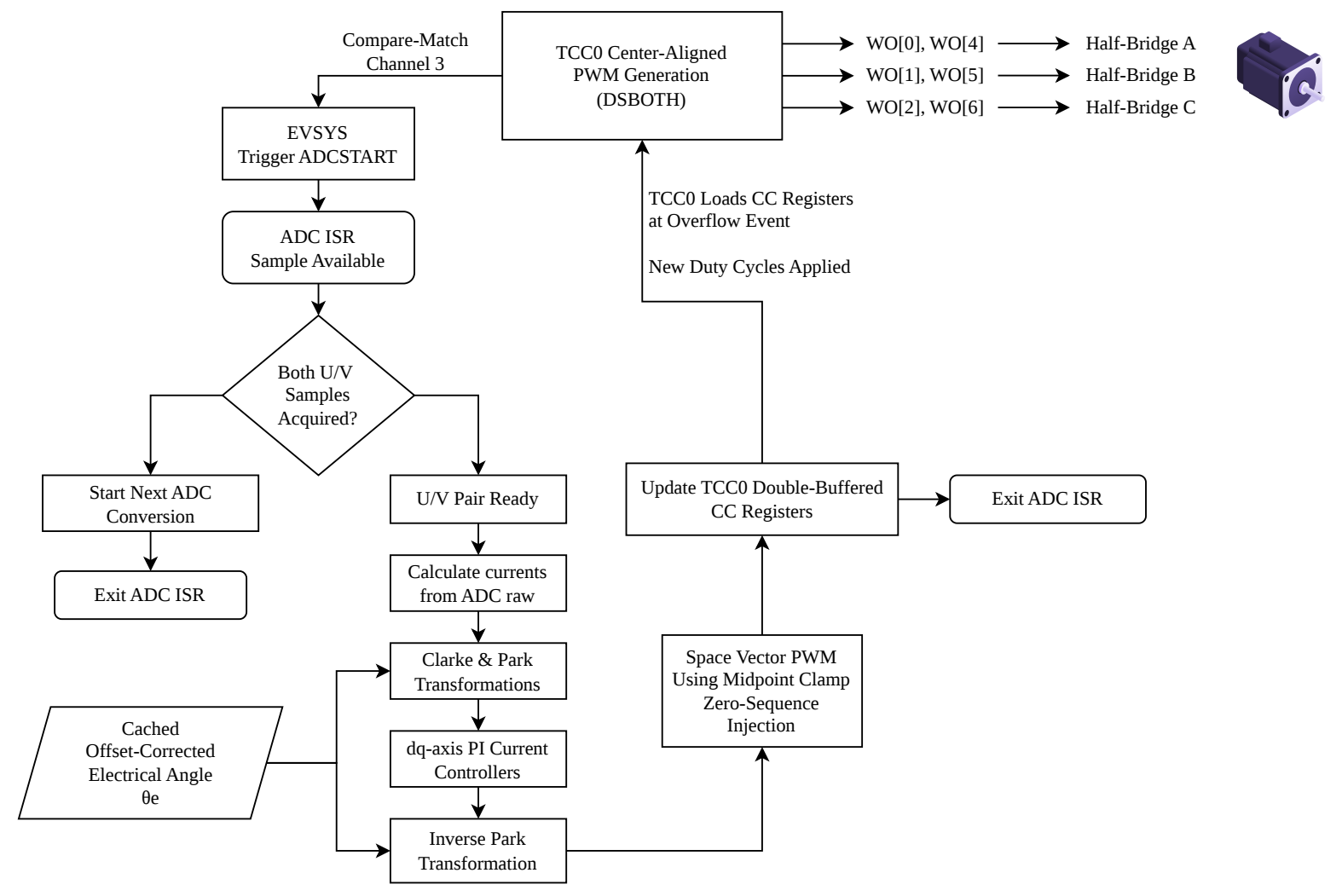
Compare-Match
Channel 3
TCC0 Center-Aligned
PWM Generation
(DSBOTH)
WO[0], WO[4]
WO[1], WO[5]
WO[2], WO[6]
Half-Bridge A
Half-Bridge B
Half-Bridge C
EVSYS
Trigger ADCSTART
ADC ISR
Sample Available
Both U/V
Samples
Acquired?
TCC0 Loads CC Registers
at Overflow Event
New Duty Cycles Applied
Start Next ADC
Conversion
Exit ADC ISR
U/V Pair Ready
Calculate currents
from ADC raw
Clarke & Park
Transformations
dq-axis PI Current
Controllers
Inverse Park
Transformation
Cached
Offset-Corrected
Electrical Angle
θe
Space Vector PWM
Using Midpoint Clamp
Zero-Sequence
Injection
Update TCC0 Double-Buffered
CC Registers
Exit ADC ISR